

Life as a Waiting Problem: A Bitcoin Analogy

Abstract

A purely peer-to-peer version of human existence would allow emotional and operational tasks to move directly from one state of life to another without requiring a centralized institution, external guarantee, or trusted authority to determine when progress has truly occurred. Traditional institutional frameworks provide part of this

solution, but much of the benefit is lost when a trusted third party, future outcome, or artificial guarantee of completion is still required before the individual can consider a matter settled.

We propose a solution to the waiting problem using a peer-to-peer model of attention and action. The system timestamps meaningful states of life by anchoring completed work to an ongoing chain of accumulated effort, forming a record that becomes increasingly difficult to contradict as subsequent work is completed. The longest chain serves not only as evidence of

the sequence of events witnessed, but as evidence of where the individual's attention and effort have actually been invested. As long as a majority of available cognitive effort remains directed toward honest progress rather than coordinated rumination, the productive chain will continue to grow and outpace competing chains of anxiety.

The internal network requires minimal structure. New concerns and pending duties are acknowledged on a best-effort basis, productive states are recorded through action, and cognitive processes may leave and

rejoin the network at will. Upon returning, the individual accepts the accumulated history of completed work as evidence of what actually happened while attention was elsewhere. The objective is not to eliminate uncertainty, but to prevent uncertainty from repeatedly spending the same finite resource: time.

1. Introduction

Life—and similarly, the psychological processing of modern existence—has come to rely heavily on institutions, schedules, promises,

and other trusted mechanisms for determining whether an outcome will eventually occur. While these systems work well enough for many ordinary milestones, they retain an inherent weakness: the individual must often continue waiting even after the relevant action has already been taken.

Completely irreversible progress is difficult to recognize while the future remains uncertain. A person may submit an application but continue checking for a response. They may complete an assignment but continue worrying about whether it was sufficient. They may

make a decision but repeatedly reopen the question because the consequences have not yet become visible. The action has occurred, yet psychologically the transaction remains pending.

The cost of waiting increases cognitive transaction costs, limiting the minimum practical amount of attention that can be devoted to ordinary tasks and reducing the ability to make small, irreversible increments of progress. With the possibility of indefinite delay, the need for reassurance spreads. Individuals become wary of their own futures and repeatedly demand

information, confirmation, prediction, and guarantees that they would not otherwise require.

A certain amount of existential friction is accepted as unavoidable. These costs and timeline uncertainties can sometimes be reduced by returning attention to the immediate present, but no universal mechanism exists for establishing mental certainty over a future-dependent communication channel.

What is needed is therefore not a system that guarantees every future outcome, but a system that allows the individual to recognize

completed progress without requiring continuous trust in the future. Tasks should be able to move directly from intention to execution, and from execution to recorded completion, without remaining psychologically unconfirmed merely because their external consequences have not yet arrived.

In this paper, we propose a peer-to-peer model of attention in which completed actions form a chronological chain of evidence. The system is secure not because anxiety can be eliminated, but because honest cognitive effort can

collectively produce more confirmed progress than competing processes can produce uncertainty.

2. Transactions

We define a unit of lived experience as a sequence of states connected by intentional action. Each owner of a period of attention transfers that attention from one operational phase to another by committing the available resource to a task and producing an observable result. The completed state references the state that preceded it and establishes the next state of activity.

A conscious mind can therefore inspect the chain of actions and determine how attention was actually spent.

The problem, however, is that the mind cannot automatically determine whether a previous phase has already consumed the same block of attention through repeated rumination. A person can spend the same two hours worrying about an examination, a job application, a conversation, or an uncertain future without producing a corresponding change in the external state.

A common solution is to introduce a trusted central authority: a calendar, institution, supervisor, notification, approval, or external event that declares when a matter is finally settled. After each period, attention is effectively returned to the authority for another confirmation. Only outcomes explicitly confirmed by the authority are treated as complete.

The problem with this solution is that the psychological fate of the entire routine becomes dependent upon the institution operating the external system. Every decision must pass through it, much like

transactions passing through a bank.

We need a way for the present mind to know that previous attention has already been allocated and that the same cognitive resource does not need to be spent again merely because the external consequence has not arrived.

For our purposes, the earliest completed action is the one that counts. Later attempts to repeatedly reconsider the same variable do not create a new productive state unless they actually change the state of affairs.

The challenge is therefore to establish a single history of meaningful action. Pending duties may be acknowledged internally, but progress is represented by what has actually been executed. The present state accepts the accumulated chain of completed work as the best available record of what has already happened.

3. Timestamp Server

The solution begins with a timestamping mechanism.

A timestamp server works by taking information that existed at a particular point and anchoring it to a chronological record. Each timestamp references the previous timestamp, creating a chain in which later records reinforce the existence and ordering of earlier records.

In our cognitive architecture, each completed block of work references the state that preceded it. The purpose is not merely to remember what happened, but to establish that something actually happened before the current state.

An unfinished thought may describe a possible future. A completed action describes a state that has already been reached.

Thus, completing a task creates a timestamp in the history of attention. Finishing an assignment, submitting an application, studying a chapter, writing a letter, making a decision, or completing a necessary conversation becomes an entry in the chain.

The emotional weight of the previous block may remain, but it no longer determines whether the work occurred. The present task is

anchored to actual labor rather than speculative possibility.

As additional blocks are added, the history becomes increasingly difficult to dismiss. The individual may still dislike the outcome of a previous action, but cannot honestly claim that no action occurred.

The chain therefore provides a distinction between **what has happened** and **what might happen next**.

4. Proof-of-Work

To implement a distributed record of progress without relying entirely on external guarantees, we require an analogue of proof-of-work.

In Bitcoin, proof-of-work requires miners to expend computational effort to produce a block satisfying a difficult mathematical condition. The resulting proof can be verified cheaply, while changing the block requires repeating the work and rebuilding the chain that follows it.

In the cognitive system, productive effort plays an analogous role.

The proof is not cryptographic.

Human attention cannot be converted directly into a SHA-256 proof, nor should it be. Instead, the analogue of proof-of-work is the irreversible evidence produced by sustained attention: the completed assignment, submitted application, finished project, learned material, repaired object, written document, or decision actually carried out.

Once genuine work has been completed, changing the historical fact that it was completed requires more than merely thinking about an alternative. The individual would have to undo or replace the result itself.

As later completed actions are chained after it, rewriting the meaning of the earlier period becomes increasingly difficult. To convince oneself that the work never happened, one must psychologically reconstruct a competing history capable of accounting for everything that happened afterward.

Proof-of-work therefore becomes proof of **investment** rather than proof of computation.

The majority decision is represented not by the loudest thought, but by

the chain containing the greatest accumulated evidence of actual attention and action. If productive effort controls the majority of available cognitive capacity, the honest chain grows faster than the competing chain of rumination.

An anxious process can attempt to reopen an old block, but it must compete against every subsequent action already built upon it.

To modify a past state, the attacker would need to reproduce the work required to establish an alternative history and then catch up with the accumulated progress of the honest

chain.

The difficulty of this cognitive reorganization increases as new productive blocks are added.

5. Network

The steps to run the internal cognitive network are as follows:

- New pending duties and concerns are acknowledged by active mental processes.
- Each active process collects relevant tasks into a current

block of attention.

- The individual works on completing the block rather than endlessly simulating its possible outcomes.
- When a meaningful result is produced, the completed block is recorded in the ongoing history of action.
- Cognitive processes accept the block only if its actions are valid, genuinely performed, and not merely repeated rumination disguised as progress.

- The mind expresses acceptance of the block by directing attention toward the next block, using the completed state as the foundation for further work.

Mental processes do not need to receive every new thought instantaneously. A concern may remain pending for a while without requiring immediate resolution.

Likewise, a person does not need to monitor every corner of consciousness continuously. Attention may leave one task and return later. Upon returning, the individual can inspect the

accumulated chain and determine what has actually been completed.

Nodes therefore do not need perfect communication. They need only enough communication for meaningful states to eventually become incorporated into the chain of action.

If several possible interpretations of a situation appear simultaneously, subsequent action determines which history receives further work. The chain that continues to accumulate genuine progress becomes the authoritative operational history.

6. Incentive

By convention, the first transaction in a productive block represents the creation of new clarity owned by the creator of the block.

This serves as an incentive to support the network because productive action does not merely complete an external duty; it also returns a portion of cognitive capacity to the individual.

The steady creation of new clarity is analogous to the issuance of new

bitcoins through mining. In Bitcoin, miners expend computational resources to add valid blocks and receive a reward for supporting the network.

In the cognitive model, the resource expended is sustained attention, while the reward is progress, competence, reduced uncertainty, or recovered mental capacity.

Task completion may also carry an internal transaction fee. Difficult tasks require greater attention, and some tasks impose costs before they generate their corresponding returns.

The incentive helps encourage cognitive nodes to remain aligned with the rules of the network. An anxious process capable of consuming a large amount of attention must choose between spending that capacity on repeatedly reconstructing catastrophic possibilities or directing the same capacity toward producing new information and completed work.

The productive chain therefore has a natural economic advantage: action produces additional evidence of progress, while rumination

frequently produces another copy of the same uncertainty.

The objective is not to make anxiety impossible. It is to make productive attention the more rewarding place for scarce cognitive resources to accumulate.

7. Reclaiming Disk Space

Once the latest transaction in a life phase has been buried under enough subsequent blocks of routine and completed work, the detailed contents of earlier states need not remain in active memory.

Human memory cannot retain every transaction indefinitely. To preserve cognitive bandwidth, completed concerns must eventually be compressed.

Bitcoin uses Merkle trees so that transactions can be represented through a single root while allowing particular transactions to be verified through their branches. This permits old transaction data to be pruned while preserving the structure needed to verify the chain.

The cognitive analogue is consolidation.

A completed experience can be reduced from an active problem to a compact memory: “that happened; I handled it.”

The individual does not need to replay every argument, worry, decision, or minute of effort associated with the event. The experience remains part of the chain without occupying the same amount of active attention.

Old worries may therefore be compacted into lessons, records, memories, or simple acknowledgements.

The objective is not to erase history. It is to prevent history from consuming the storage required to continue living.

8. Simplified Payment Verification

It is possible to verify life progress without running a full and exhaustive psychological node.

An individual does not need to reconstruct every thought ever experienced before beginning the next task. Instead, they need only

enough evidence of the longest chain of completed work to establish that the present state follows from previous states.

A person returning after a period of absence can inspect the block headers of their recent history: completed assignments, submitted applications, finished projects, conversations held, decisions made, and other meaningful milestones.

A particular task can then be linked to the block in which it was completed. Subsequent completed blocks provide additional confirmation that the event belongs

to the actual history rather than merely to an imagined future.

The longer the chain continues after an event, the less useful it becomes to reopen the question of whether that event happened.

This does not prove that the original action was correct. It proves something narrower and more useful:

the action occurred, and life continued from it.

The system therefore provides a lightweight method for verifying

progress without requiring exhaustive psychological replay.

9. Combining and Splitting Value

Although it would be possible to treat every minute of attention as an independent unit, doing so would be impractical.

Human attention must therefore be combined and divided according to the scale of the task.

A large block of accumulated experience may provide the input for

a new project. Several smaller periods of attention may combine to produce a larger result. Likewise, a large task may produce multiple outputs: one belonging to the immediate objective and another returning knowledge, experience, confidence, or remaining energy to the individual.

Normally there will be a single input from a previous project or several inputs representing smaller completed efforts, and one or more outputs representing the new state of life.

For example, hours spent studying

may combine into an examination result. Several small programming exercises may combine into a functioning application. Previous employment, study, and personal projects may combine into a new professional opportunity.

The value is therefore not in the individual hour alone, but in how previously accumulated effort is transferred into new states.

Just as Bitcoin transactions can consume multiple inputs and create multiple outputs, human progress is rarely produced from a single isolated action.

10. Privacy

The traditional institutional model achieves privacy by limiting information to the parties involved and, in some cases, the trusted third party.

A completely transparent cognitive network would obviously be unacceptable. Not every thought, fear, intention, or unfinished task should be broadcast to every other person.

Privacy can therefore be maintained

by separating the public evidence of progress from the private content of experience.

The external world may see that an individual completed a task, submitted a project, attended an examination, or reached a milestone without seeing every private thought that preceded it.

The chain records the existence and ordering of meaningful actions without requiring the publication of every internal process that produced them.

A person may therefore maintain

many private cognitive states while exposing only the minimum information necessary to establish the validity of completed work.

The purpose of privacy is not to conceal the chain of progress, but to prevent the contents of the mind from becoming another source of unnecessary external pressure.

The individual remains free to generate new cognitive focuses for different phases of life, preventing every task from becoming permanently linked to every previous fear, failure, or uncertainty.

11. Calculations

We consider the scenario of an anxious attacker attempting to generate an alternate chain of catastrophic interpretations faster than the honest chain of productive action.

Even if this is accomplished temporarily, it does not give the attacker arbitrary control over reality. Anxiety cannot create a failure that never occurred, nor can it manufacture a lost opportunity merely by imagining one.

A thought is not automatically a valid transaction.

Honest cognitive processes reject invalid states by refusing to treat speculation as evidence. An imagined disaster does not become part of the historical chain simply because it was simulated repeatedly.

The attacker can therefore attempt primarily to reorganize the individual's interpretation of previous events or reclaim attention that has already been allocated to productive work.

The race between the honest chain

of action and the attacker's chain of rumination can be characterized as a binomial random walk.

The success event is the honest chain being extended by another block of productive work, increasing its lead by one. The failure event is the anxiety chain being extended by another block of rumination, reducing the gap.

Given that productive effort controls the greater share of available cognitive resources, the probability of the competing chain catching up decreases as the honest chain grows.

This is analogous to Bitcoin's security argument: modifying an older block becomes increasingly difficult because an attacker must reproduce the work represented by that block and every subsequent block, then overtake the honest chain.

In human terms, the longer a person continues to build a real history of completed action, the more expensive it becomes to replace that history with a purely hypothetical one.

A person who spends one evening

worrying may still be able to reverse course the next morning. A person who has spent months learning, working, building relationships, submitting applications, completing projects, and accumulating evidence of progress has a much larger chain behind them.

The anxious attacker can still produce blocks.

It simply has to work harder to make those blocks matter more than the history of everything that has actually been done.

12. Conclusion

We have proposed a model of life as a waiting problem in which scarce human attention is treated as a resource that can either accumulate into productive history or be repeatedly consumed by uncertainty.

We began with the familiar problem of tasks that remain psychologically pending even after the individual's part has been completed.

Traditional institutions can confirm outcomes, but dependence upon confirmation can create a persistent state of waiting in which the same

block of attention is repeatedly spent.

To address this, we proposed a peer-to-peer model in which meaningful actions are recorded as sequential states and strengthened by subsequent work.

The system does not require perfect certainty about the future. It requires only a reliable distinction between what has happened and what has not yet happened.

Proof-of-work becomes an analogue for accumulated effort: completed work provides evidence that

attention has actually been invested. The chain becomes a history of lived action. Confirmation becomes the passage of subsequent completed states. Double-spending becomes the repeated consumption of the same finite attention by rumination. The attacker becomes the competing process that attempts to replace accumulated action with hypothetical outcomes.

The internal network remains robust because it does not require every thought to be resolved immediately. Pending transactions can remain pending. Nodes may leave and

return. Attention may move between tasks. Old blocks may be compressed into memory. What matters is that the chain of actual action continues to grow.

The central principle is therefore simple:

A future outcome does not need to be confirmed before the work required of the present can be considered complete.

The chain cannot guarantee that the next block will be favorable. It can only prove that the previous blocks were built.

And perhaps that is the solution to the waiting problem.

Not certainty.

Not prediction.

Not the elimination of anxiety.

But a history strong enough to remind the present that it has already done its work.

The future may remain unconfirmed.

The block is still there.

